Individual member contributions

Project: Learning When to Search for Multi Agent Path Finding
Institution: IIT Roorkee
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Allocated role: Problem formulation, learned policy and search methods

The project idea and the central question of when a learned MAPF policy should invoke search are shared contributions. My individual allocation covers the environment, movement policy and search components, together with the corresponding technical explanation and submission review.

1. Problem formulation and environment

Responsibility for the MAPF formulation: starts and goals, legal actions, vertex conflicts, edge-swap conflicts, flowtime and makespan. This scope includes reviewing the random-grid environment and BFS distance fields, and explaining the distinction between the full 32×32 map and an agent's 11×11 observation window.

2. Learned policy and imitation learning

Responsibility for the policy methodology: four observation channels, CNN action probabilities, invalid-action masking and imitation learning from LaCAM trajectories. Review coverage includes the training procedure, the 74.6% held-out action agreement, the sample-based validation split and why action accuracy alone does not establish episode success.

3. Collision shielding and search components

Responsibility for explaining the collision shield, priority inheritance with backtracking, policy-guided CS-PIBT and full-horizon LaCAM. The allocated technical review covers candidate ordering, sampled tie-breaking, collision safety, and the distinction between one-step coordination and planning across several timesteps.

4. Method documentation and code walkthrough

Responsibility for the method sections of the report and the first presentation segment. This includes the system pipeline, policy training, search hierarchy, algorithm explanations and a walkthrough of `grid.py`, `policy.py`, `pibt.py` and `lacam.py`.

5. Integration and submission review

Responsibility for reviewing how the environment, policy and search modules connect to the controller; checking setup instructions and model loading; and coordinating with Shivang on the consistency of the method description and experiment interpretation.

Primary deliverable scope: Problem formulation, policy/search methodology, method presentation and technical code explanation. These responsibilities are the agreed allocation for individual preparation and review.